Mohit Vaid

The George Washington University | [website] | [m.vaid@gwu.edu] | [Linkedln]| | [Github]

RESEARCH INTEREST

Theoretical and applied cryptography; recent focus on Functional Encryption, Distributed Point Functions, Private Set Intersection, Side-Channel Attacks on Post-Quantum Cryptographic Schemes, Differential Privacy

EDUCATION

The George Washington University — **Washington, DC**
Doctor of Philosophy, Cryptography — August 2024 - Present

- Relevant Coursework: Design and Analysis of Algorithms, Introduction to Cryptography, Differential Privacy, Advanced Software Paradigms

The Johns Hopkins University — **Baltimore, MD**
Master of Science — August 2023-May 2024

- Relevant Coursework: Practical Cryptography, Security and Privacy in Computing, Intrusion Detection, Intro to Algorithms

Indian Institute of Technology Kanpur — **Kanpur, India**
Master of Technology — August 2020-June 2022

- Relevant Coursework: Numerical Methods, Probability and statistics

National Institute of Technology Kurukshetra — **Kurukshetra, India**
Bachelor of Technology — July 2012-June 2016

- Relevant Coursework: C Programming, Engineering Mathematics - I , II , III

PRESENT PROJECTS

Updatable Private Set Intersection — **Oct 2025 - present**

- Developing multi-party UPSI protocol with succinct communication and computation complexity

Rowhammer attacks on Post Quantum Secure Cryptosystems — **Oct 2025 - present**

- Developing techniques to explore attacks on post quantum secure McEliece KEM under guidance of Prof Arkady Yerukhimovich and Prof Daniel Genkin (Georgia Tech).

Functional Encryption — **June 2025- present**

- Developing a single-input functional encryption scheme for the inner-product function class with improved computational efficiency under guidance of Prof Arkady Yerukhimovich and Prof Adam O'Neill.

Distributed Point Function — **August 2024 - present**

- Developing protocol for multiparty distributed point function for dishonest majority under the Supervision of Prof Arkady Yerukhimovich.

PAST PROJECTS

Functional Encryption — **June 2025- Dec 2025**

- Developed multi-client functional encryption for restricted function relying on minimal cryptographic assumption under supervision of Prof Arkady Yerukhimovich. [link]
- detect cheating behavior during online interviews, featuring adjustable detection thresholds and automated session logging. [link]

Blockchain Based Ransomware Proof-of-Concept — **January 2024-May 2024**

- Leveraged Ethereum smart contracts to build an automated escrow service for fair exchange mechanism whilst exploring technical feasibility and security implications under the supervision of Prof Mathew Green at the Johns Hopkins University. [link]

Padding Oracle attack on E2E Messaging Client **January 2024- May 2024**

- Developed with one other an End to End Messaging Client based on ECDH, ChaCha20 and ECDSA, and conducted padding oracle attacks to study the security risks for the client, under Prof Mathew Green at Johns Hopkins University. link

Real Time Proctoring Application **January 2025 - May 2025**

- Developed a Python-based proctoring system compatible with any video conferencing platform leveraging MediaPipe Face Mesh and OpenCV for real-time blink rate and head movement analysis to

WORK EXPERIENCE

Dunnhumby India Limited **Gurugram, India**

Junior Developer July 2022 -July 2023

- Developed UI and API based upon react and C# respectively for displaying the real time Jobs and also the options to cancel, delete or modify Jobs

HSCC India Limited **Noida, India**

Project Management July 2016 - July 2019

PUBLICATIONS

Multi-Client Functional Encryption over Small Domains **Eprint**

Co-Authors: Arkady Yerukhimovich, Suvasree Biswas

- We construct a Private-Key Multi Client Functional Encryption secure for a bounded number of key and encryption queries based upon minimal assumption of one-way Function. link

TEACHING EXPERIENCE

Discrete Structures-1, GWU Washington DC Jan 2026-present

Design and Analysis of Algorithms, GWU Washington DC August 2025-December2025

Numerical Methods , IIT Kanpur India August 2021-December 2021

CONFERENCES/WORKSHOPS ATTENDED

Secure Computation Workshop at Simon’s Institute 2025 Berkeley, USA

PETS 2025 Washington DC, USA

Indocrypt 2024 Chennai, India